
Design and Control of an 3DOF Automated Robotic Arm System

Mechanical Design, Electrical Architecture, and Embedded
Control Implementation

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1 Introduction

1.1 Project Overview

This document presents the design, fabrication, and control implementation of a 3 Degrees of Freedom (3DOF) automated robotic arm equipped with custom end effector attachments. Designed for precise pick-and-place operations or workplace lighting needs, the system mimics basic structure of a human arm (base rotation, shoulder pitch, and elbow pitch) to manipulate objects within a defined workspace. The project integrates mechanical engineering, electrical design, and embedded systems to create a fully functional automated manipulator.

1.2 Motivation and Objective

The primary objective of this project was to develop a modular robotic system capable of demonstrating fundamental principles of robotics, including forward and inverse kinematics, servo control, and serial communication. Unlike pre-built kits, this arm was designed from the ground up over several months to allow for

- **Custom Kinematics:** Optimization of link lengths and joint ranges for specific tasks
- **Integrated Control:** Direct Manipulation via a Raspberry Pi Pico microcontroller using serial commands
- **Validated Design:** Verification of mechanical integrity and workspace reach through Matlab simulations prior to fabrications

1.3 System Architecture

1. **Mechanical Structure:** Designed using SolidWorks, the arm features lightweight links and custom designed modular attachments for secure object handling or proper lighting for your workspace.
2. **Electrical System:** Power distribution and signal routing are managed through a custom schematic designed in KiCad. The system utilizes 3 high torque servo motors and 1 Nema 17 stepper motor controlled by one A4988 stepper driver which is all driven by PWM signals given from the Raspberry Pi Pico.
3. **Control Logic:** The firmware interprets serial commands from a host computer to calculate joint angles. Control Algorithms were first modeled in Matlab to ensure smooth trajectory planning.

2 System Specifications

2.1 Performance Metrics

Parameter	Specification	Notes
Degrees of Freedom	3	Base, Shoulder, Elbow
End-Effector	Gripper and Ring Light	Actuated by DS3218 Servo and standard ring light
Controller	Raspberry Pi Pico	RP2040 Chipset
Communication	UART Serial	Baud Rate 9600
Power Supply	Ovonic 3S Lipo Battery 2200mAh	Paired with an 8A UBEC 5V / 6V for voltage step down
Max Reach	19 inches	From base center
Payload Capacity	830g	At full extension
Materials	PLA	Main structural components

2.2 Payload Capacity Analysis

The payload limit of 830 grams was derived through static torque analysis of the shoulder joint which, experiences the highest moment load and calculations of the 2nd link which would possibly be a bottle neck.

- **Calculation Method:** The maximum torque was calculated using the formula

$$\tau = m \cdot g \cdot r$$

where m is the total mass (payload + end-effector + forearm), g is gravity, and r is the distance from the joint to the center of mass

- **Safety Margin:** A safety factor of 1.5 was applied to the theoretical stall torque of the DS3218 Servos to prevent gear stripping and ensure longevity under dynamic loading (acceleration/deceleration).
- **Trade-off:** Increasing the payload capacity would require either higher-torque (and heavier) servos, which would paradoxically reduce the net payload due to increased arm weight, or shortening the arm links to reduce the moment arm.

2.3 Reach vs. Payload Trade-off

The 19-inch reach was selected to maximize the workspace envelope while maintaining structural rigidity. Originally there was a custom made 10 to 1 gear box attached to the shoulder joint to maximize potential payload, but the issue was that we had limited time and limited resources as continuous high torque servo motors are quite expensive. So we settled on 40 kg * cm servo motors for the shoulder and elbow joint, as we felt this provided a balance between being useful for a workspace (e.g electronics space, workbench etc) and being cost efficient.

- **Moment Arm Impact:** As the arm extends, the torque required at the base and shoulder joints increase linearly. At full extension (19-inches), the system operates near its maximum rated torque, limiting the payload to 830g.
- **Deflection Control:** Extending the reach beyond 19 inches with the current servo selection would result in excessive mechanical deflection (sag) at the elbow joint, compromising positional accuracy.
- **Optimization:** The link lengths were iteratively optimized in Matlab to find the “sweet spot” where the workspace volume is maximized without exceeding the torque limits of the DS3240MG which are our 40 kg * cm servo motors.

2.4 Component Selection

- **Servos:** The DS3240MG was chosen as previously stated for its high torque-to-weight ratio (40kg * cm) and its cost effectiveness, this motor is paired with the DS3218 servo motor (20kg * cm) for properly handling our payloads (tools, electrical components, lights, wires, etc) without backlash.
- **Structure:** Links were fabricated from PLA with a Bamboo A1 3D printer to minimize inertial load on the base motor while maintaining not only sufficient stiffness but also minimizing costs.
- **Control:** The Raspberry Pi Pico was selected for its dual-core architecture, allowing one core to handle serial communication while the other manages PWM timing for jitter free servo control. This micro controller was selected after extensive testing with the Arduion uno R3 paired with a PCA9685 board.

2.5 Matlab Figures

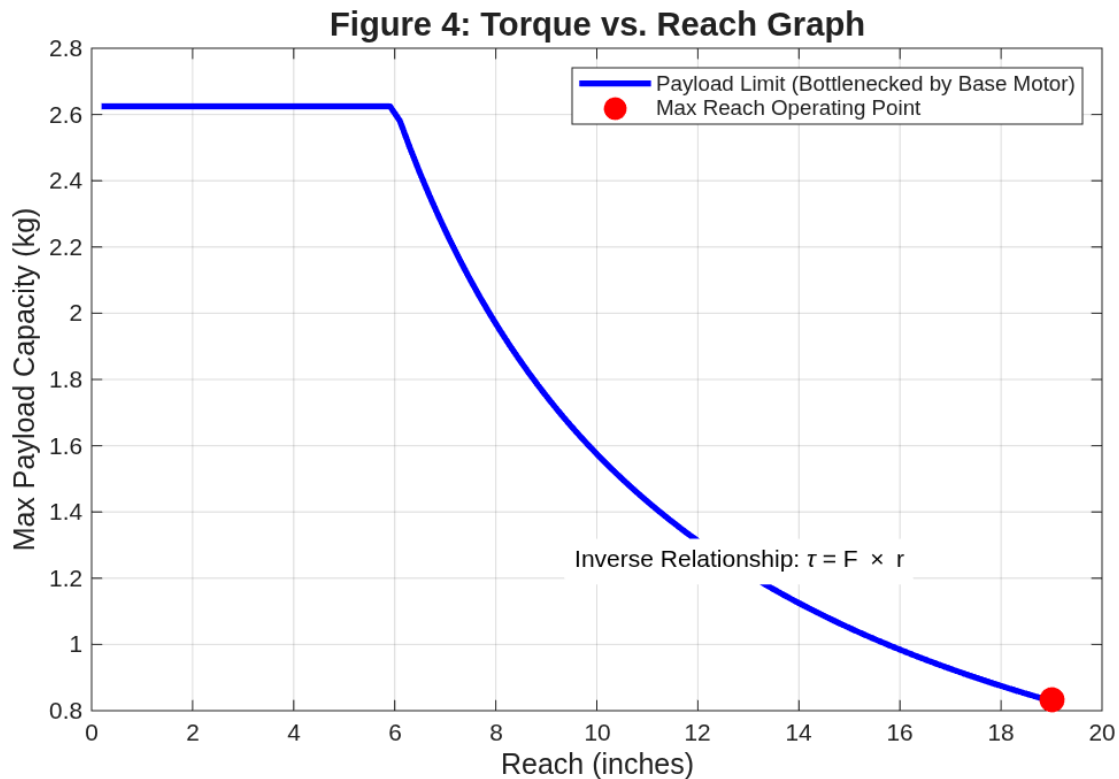


Figure 1: Payload Capacity vs. Horizontal Reach. Simulation results demonstrating the inverse relationship between the arm's horizontal extension and its maximum lifting capacity. The curve is bounded by the base motor's stall torque (40 kg·cm), resulting in a maximum payload of 0.83 kg at full extension (19 inches). The sharp decline highlights the base joint as the primary bottleneck for this configuration.

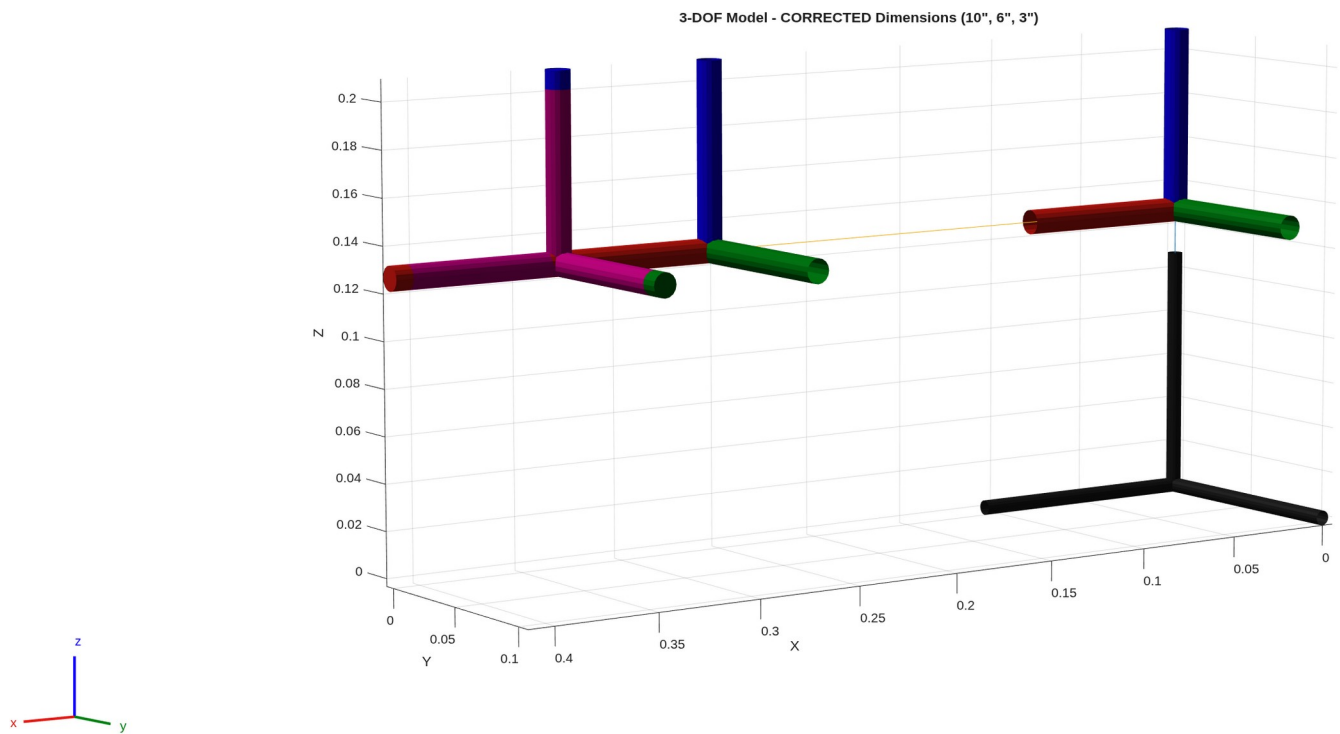


Figure 2: 3-DOF Robot Arm Kinematic Model. Visualization of the serial manipulator created in MATLAB Robotics System Toolbox. The model consists of a rotating base (Joint 1), shoulder pitch (Joint 2), and elbow pitch (Joint 3) with link lengths of 10, 6, and 3 inches respectively. The green frames indicate the local coordinate systems used for inverse kinematics calculations.

3 Mechanical Design

3.1 CAD Modeling

The mechanical assembly was designed in SolidWorks to optimize weight-to-strength ratios. The design process involved iterative stress analysis to ensure the arm could support the target payload without excessive deflection.

- **Base Assembly:** Provides stable rotation using a Nema 17 stepper motor mounted into a rectangular chassis made of PLA.
- **Arm Links:** Two links (Upper Arm and Forearm) were designed with internal channel to route servo wiring cleanly.
- **End-Effector:** A symmetric crab-pincer mechanism was selected to minimize complexity and maximize grip along with a ring light attachment for lighting assistance in various workspaces.

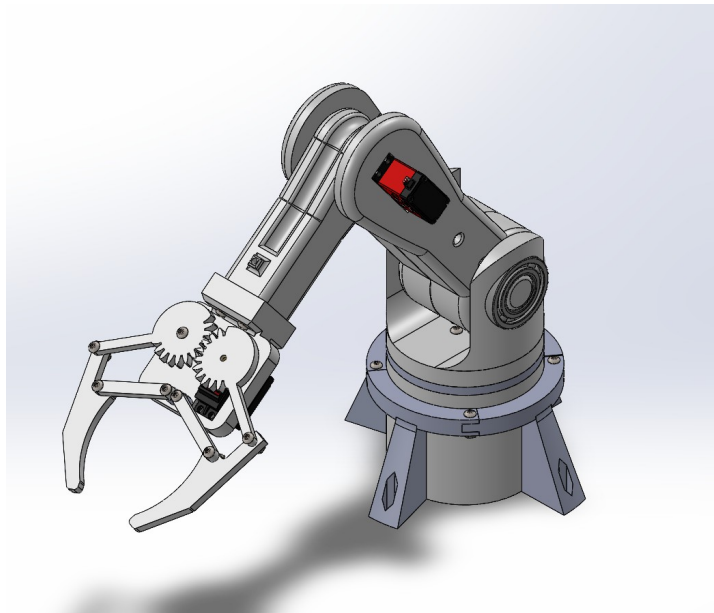


Figure 1: 3D Reachable Workspace Envelope. Full assembly view showing joint ranges and mounting points.

3.2 Kinematic Simulation

Before manufacturing, the design was validated using Matlab. Forward and inverse kinematics equations were derived to map Cartesian coordinates (X, Y, Z) to servo angles (θ_1 , θ_2 , θ_3).

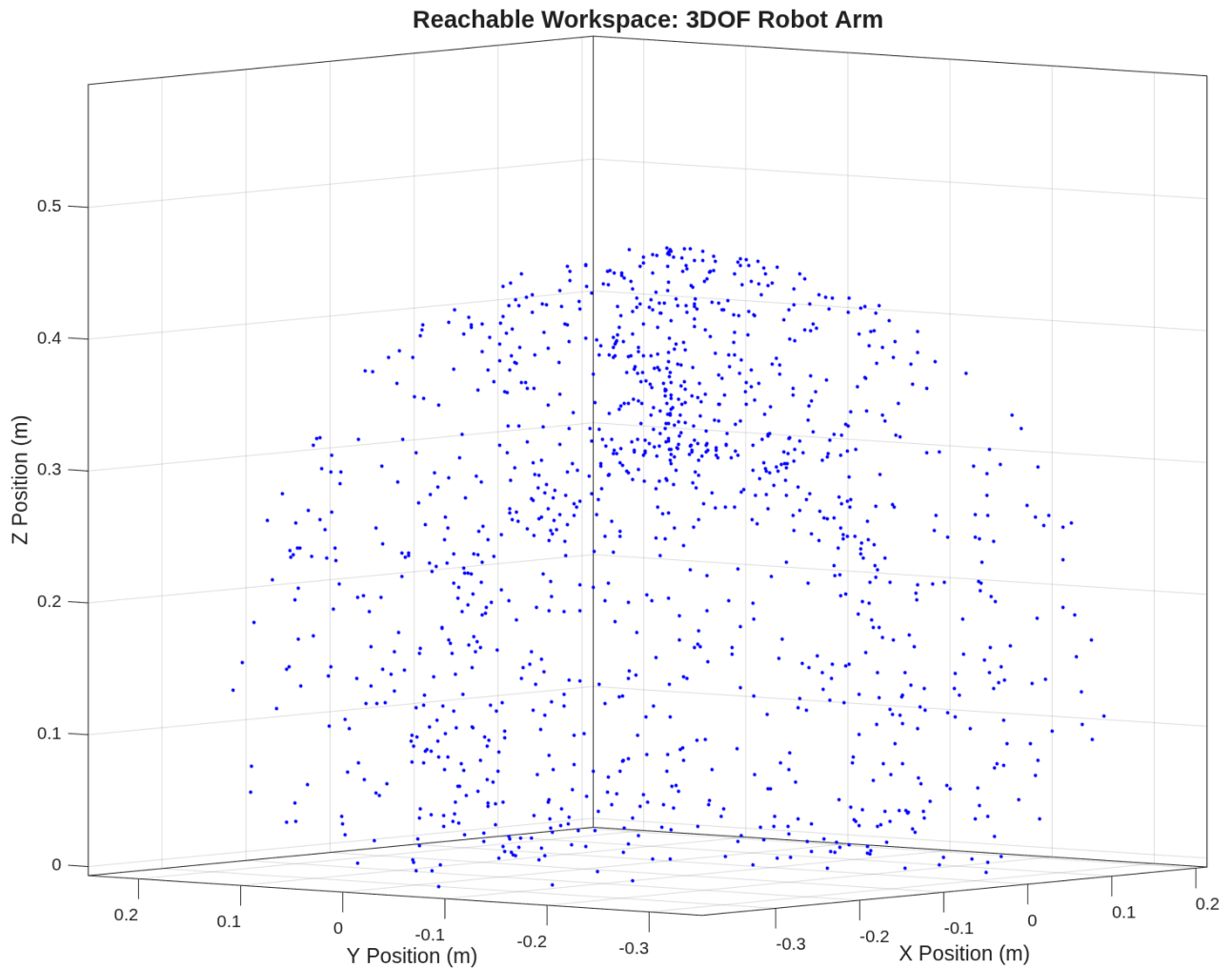


Figure 2: 3D Reachable Workspace Envelope. Matlab simulation showing the reachable workspace envelope.

3.3 Manufacturing Assembly Drafts

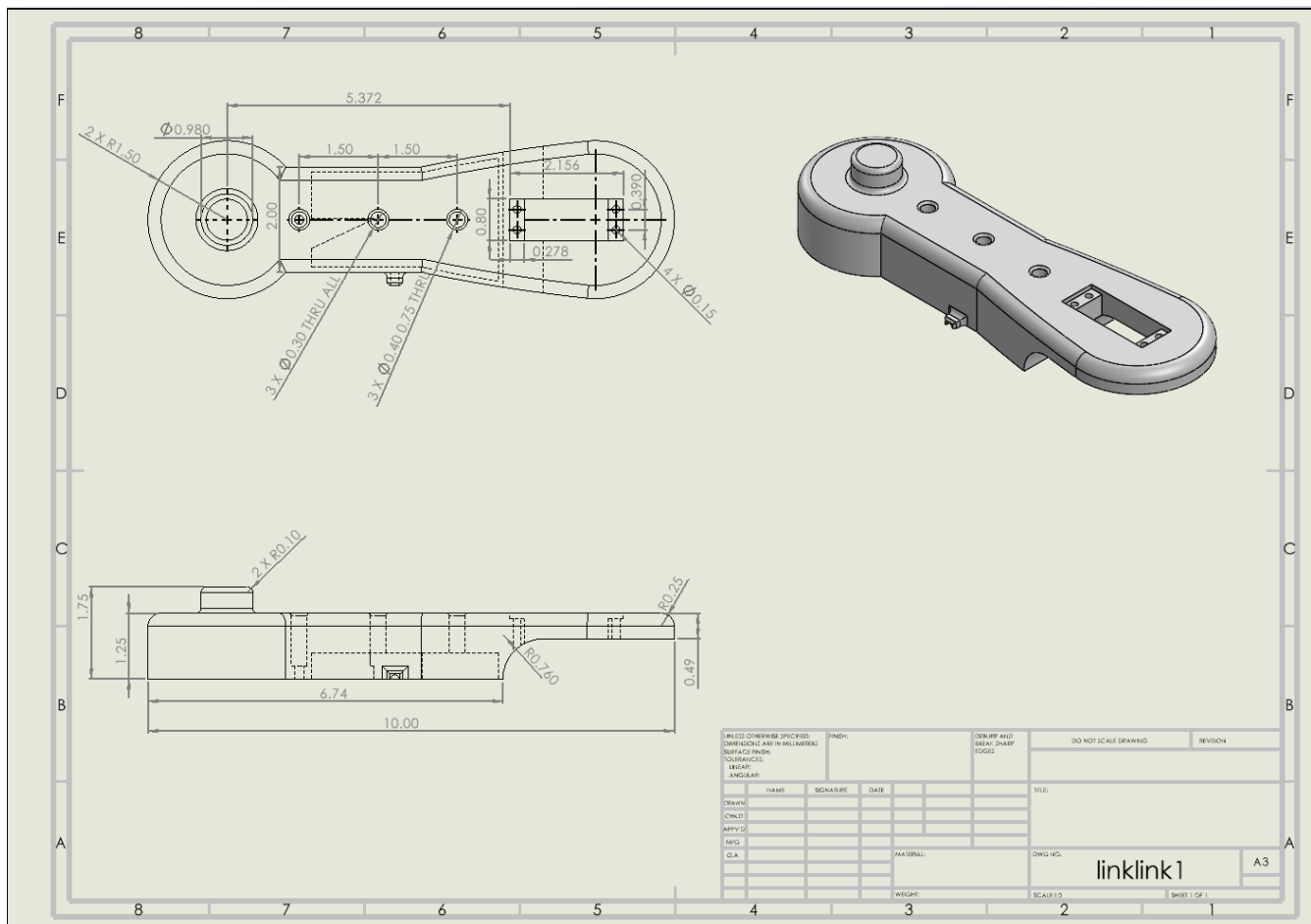


Figure 3: Link 1 (Base Arm). Detailed orthographic projections for the split-housing design of the 10-inch base link. (A) shows the primary half-shell featuring the servo mounting recess and alignment pin holes; (B) shows the mating half-shell. Note the symmetrical bolt patterns along the parting line intended for M3 fasteners. Dimensions are in inches unless otherwise specified. Second half on next page.

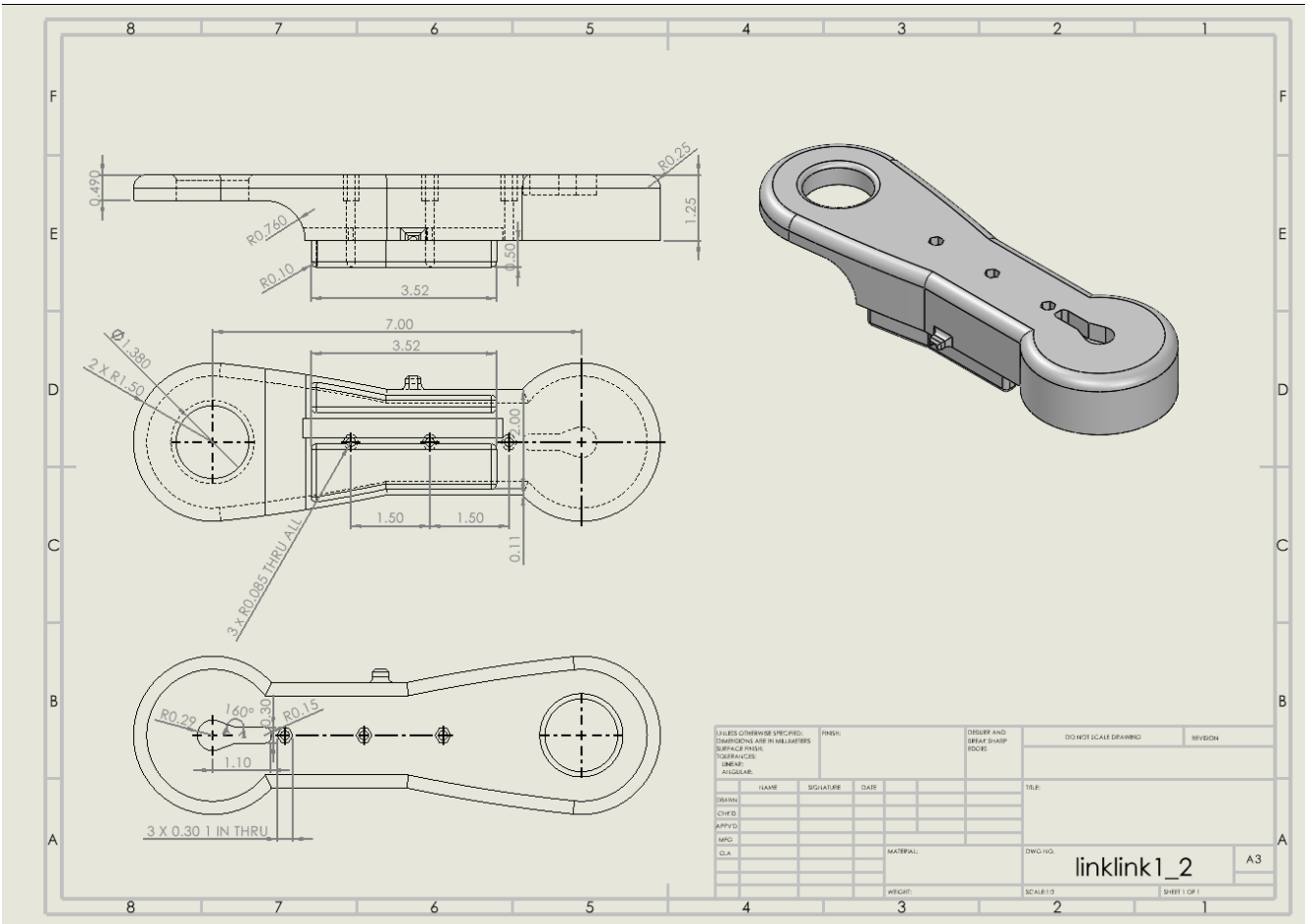


Figure 4: Link 1 (Base Arm)Second half of link 1

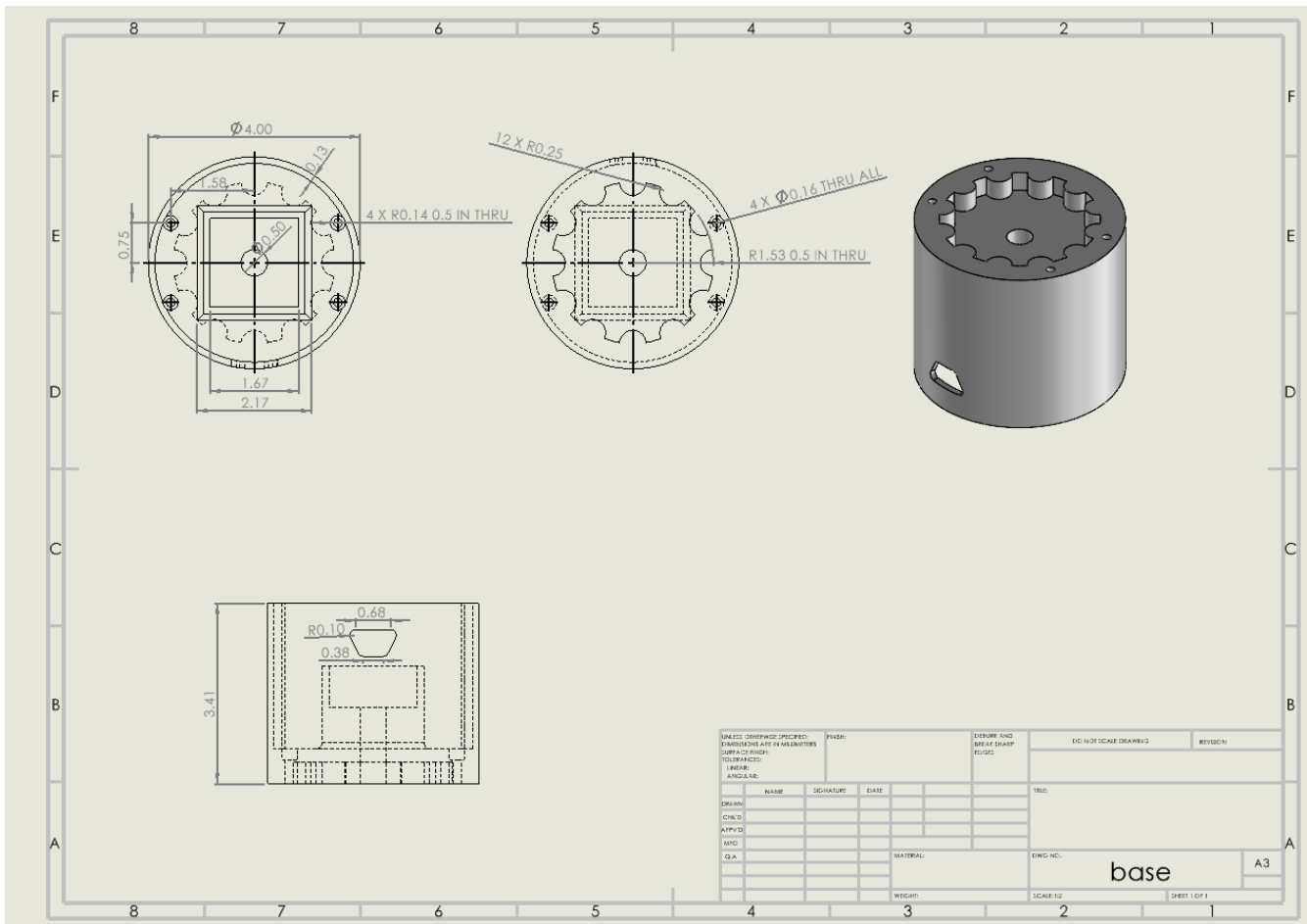


Figure 5: Servo Gearbox Housing. Exploded view of the custom 3D-printed enclosure designed to house the Nema 17 motor. The housing features integrated mounting points for the base chassis and a sealed aperture for the output shaft to minimize dust ingress while maximizing torque transmission efficiency.

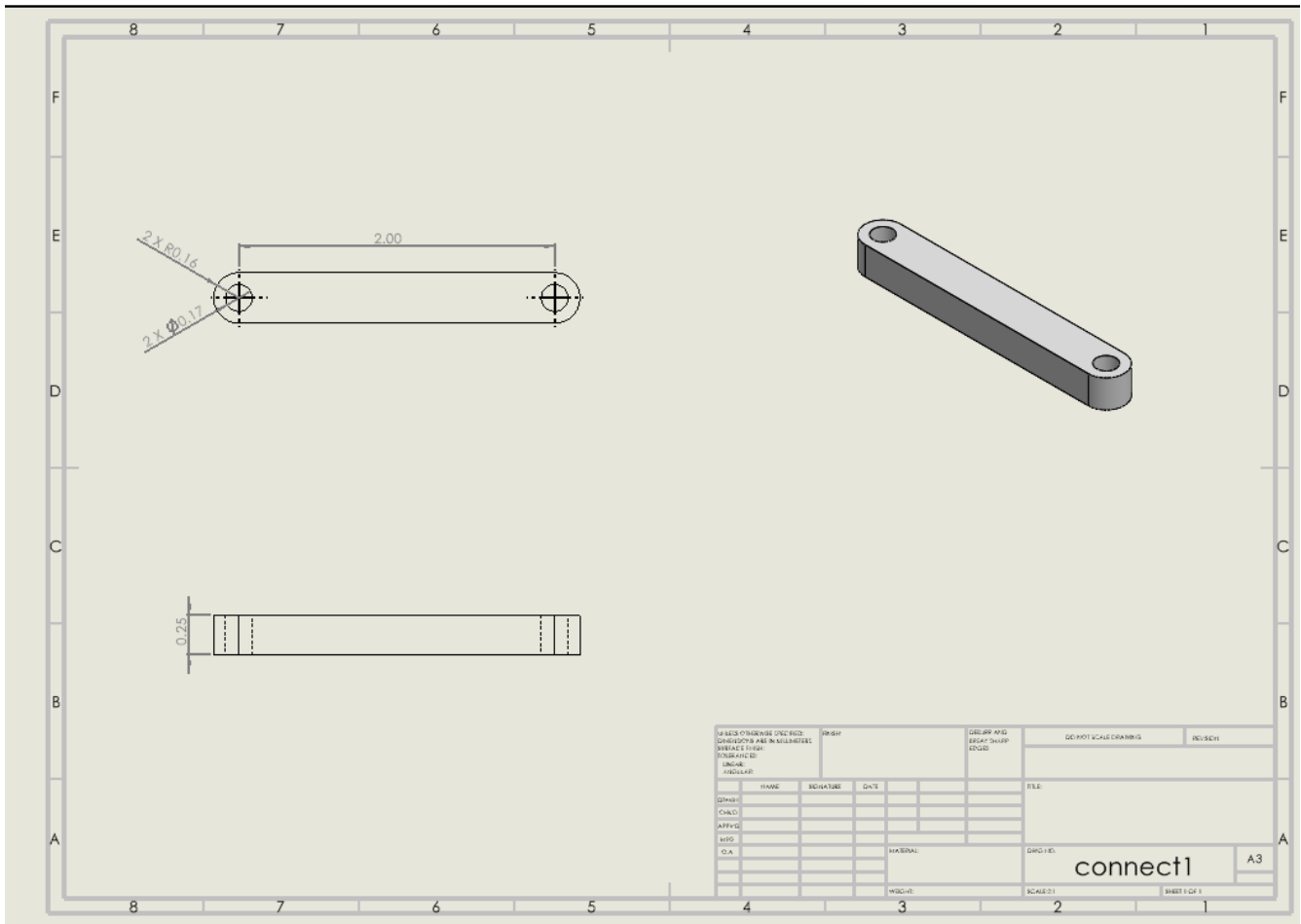


Figure 6: Drive Linkage Assembly. The mechanical interface connecting the servo gearbox output shaft to the pincher jaws. This linkage converts the rotary motion of the servo into the linear closing action of the gripper. The design utilizes a bolted connection to eliminate backlash and ensure precise synchronization between the motor angle and jaw position.

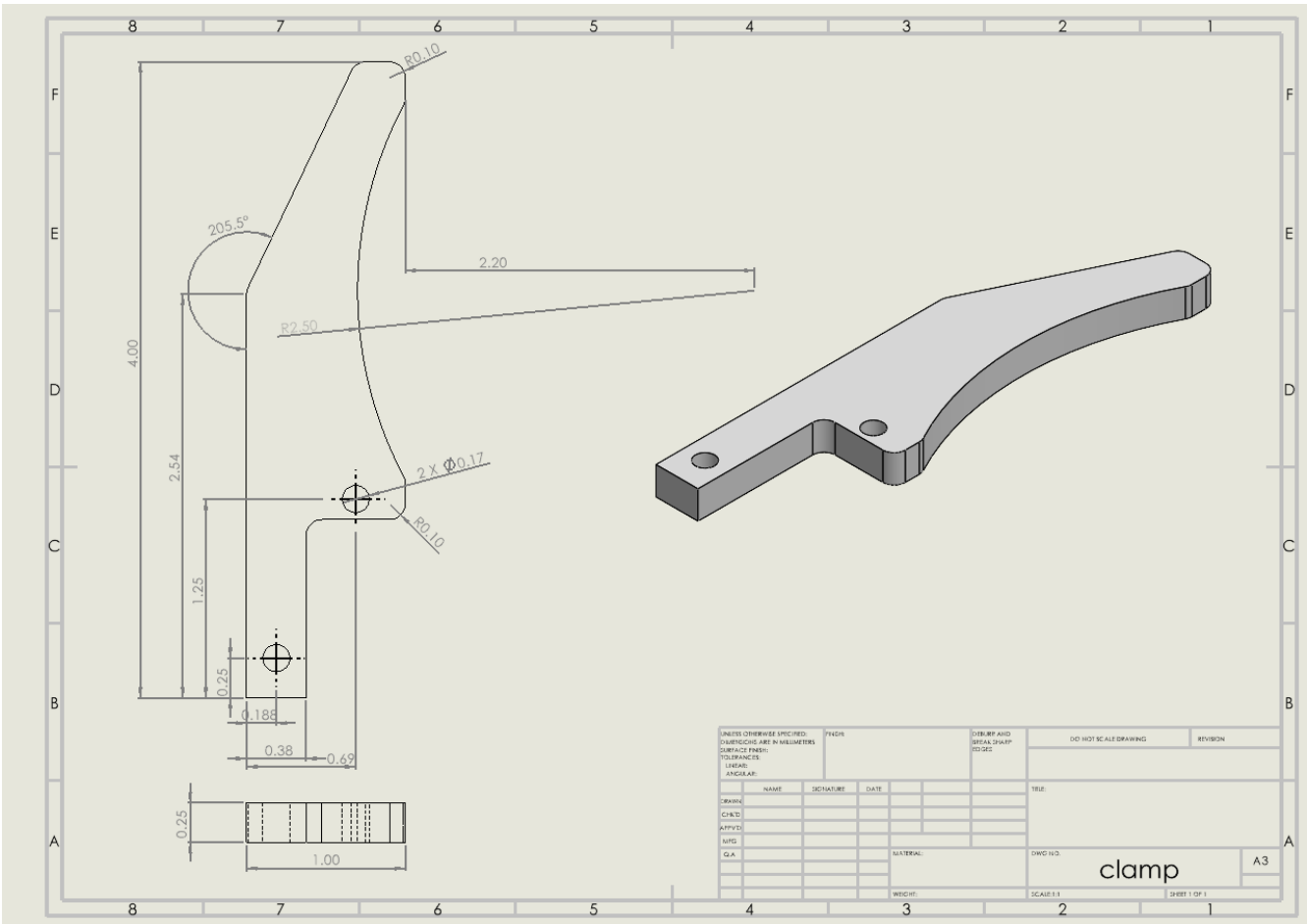


Figure 7: Crab-Style Pincher Jaws. Detailed view of the symmetric end-effector jaws. The geometry is optimized to distribute grip force evenly across the contact surface.

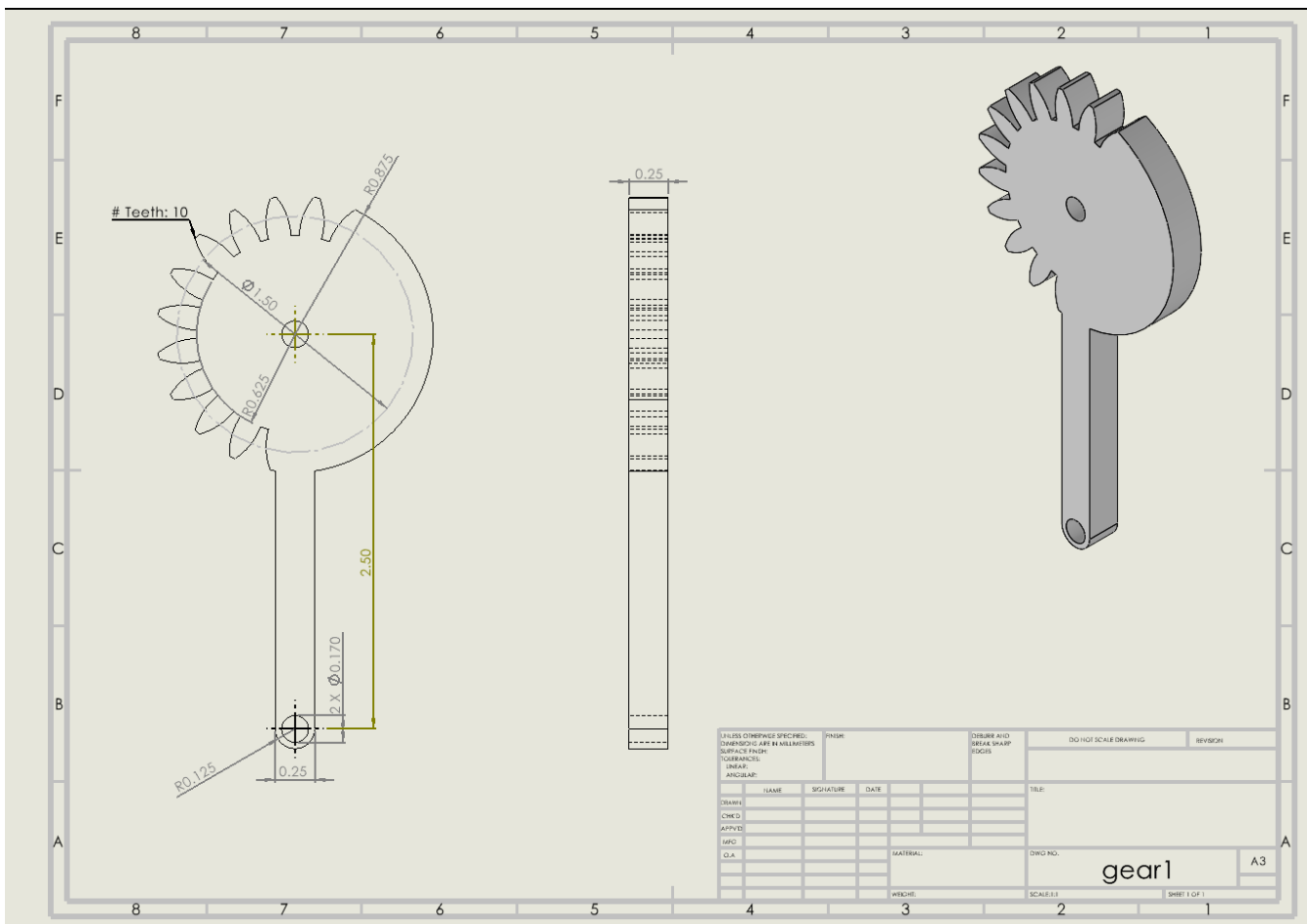


Figure 8: Gripper Drive Gear Manufacturing Draft. Detail drawing of the custom spur gear designed to couple the 20 kg·cm servo to the gripper mechanism. The caption specifies the module (pitch), pressure angle, and tooth count required to achieve the desired gear ratio. Keyway dimensions are provided to ensure zero-slip torque transmission from the servo spline to the gear.

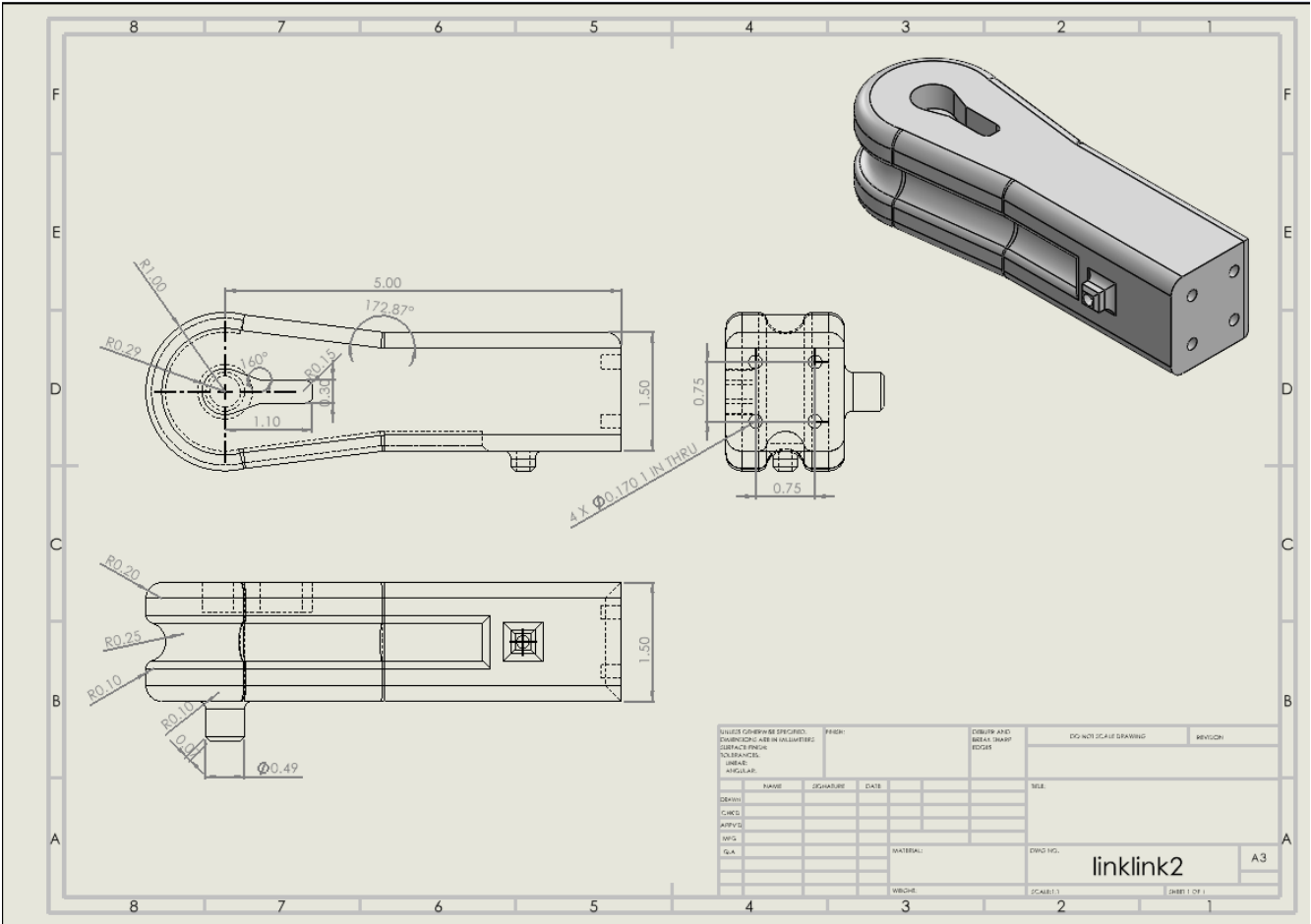


Figure 9: Link 2 (Forearm) Manufacturing Draft. Complete dimensional drawing of the 6-inch intermediate link. The draft highlights the central cutout for weight reduction while maintaining structural rigidity around the high-stress servo horn mounting points. Tolerances for the pivot bores are specified at H7 to ensure smooth rotation with standard bearings

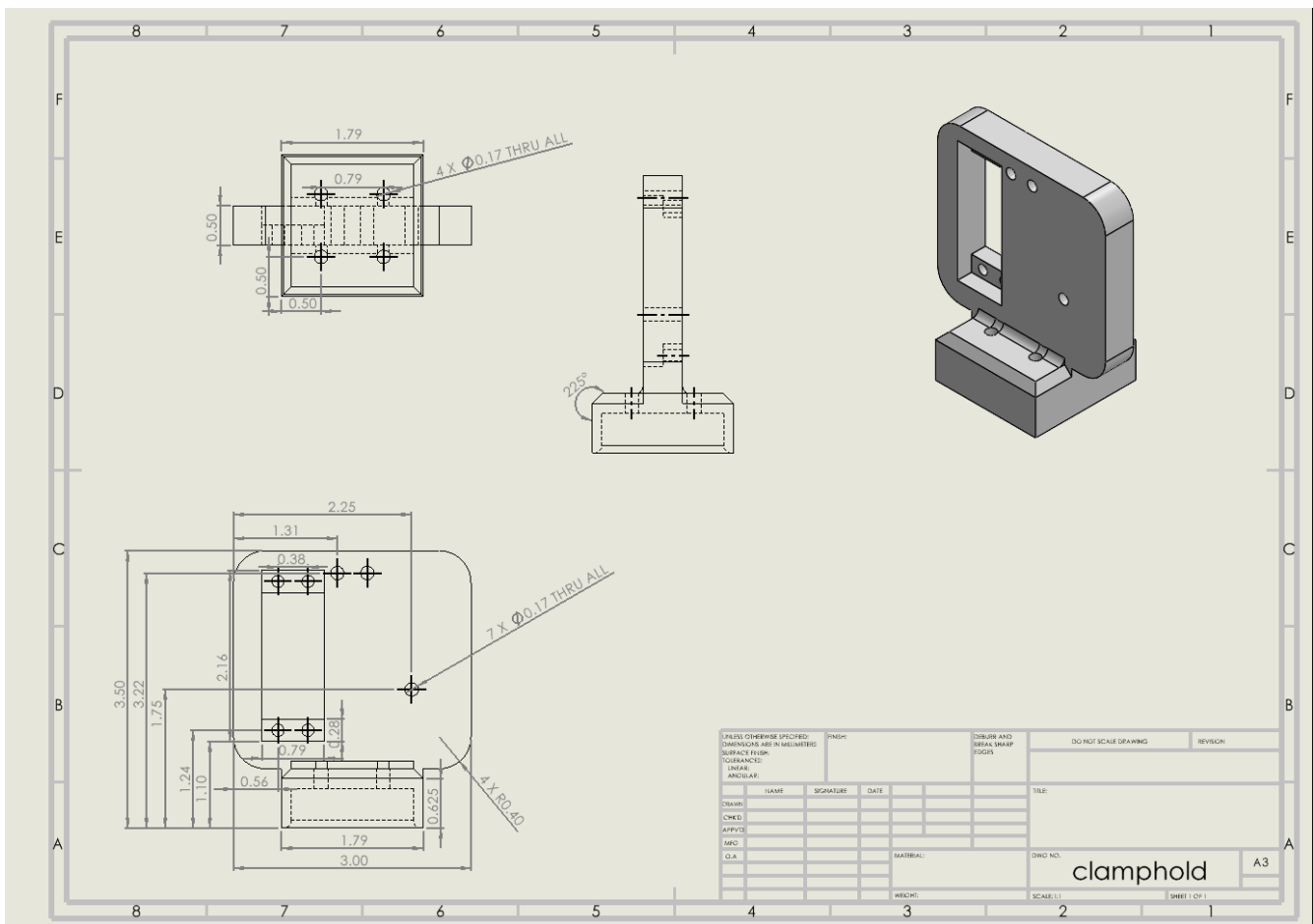


Figure 10: Gripper Assembly Housing and Connector Draft. Detailed orthographic view of the integrated end-effector mount designed to house the 20 kg·cm grabbing servo, reduction gearing, and pinchers. The drawing specifies critical mating surfaces for the Link 2 connection are dimensioned to ensure precise torque transmission and structural rigidity during grasping operations.

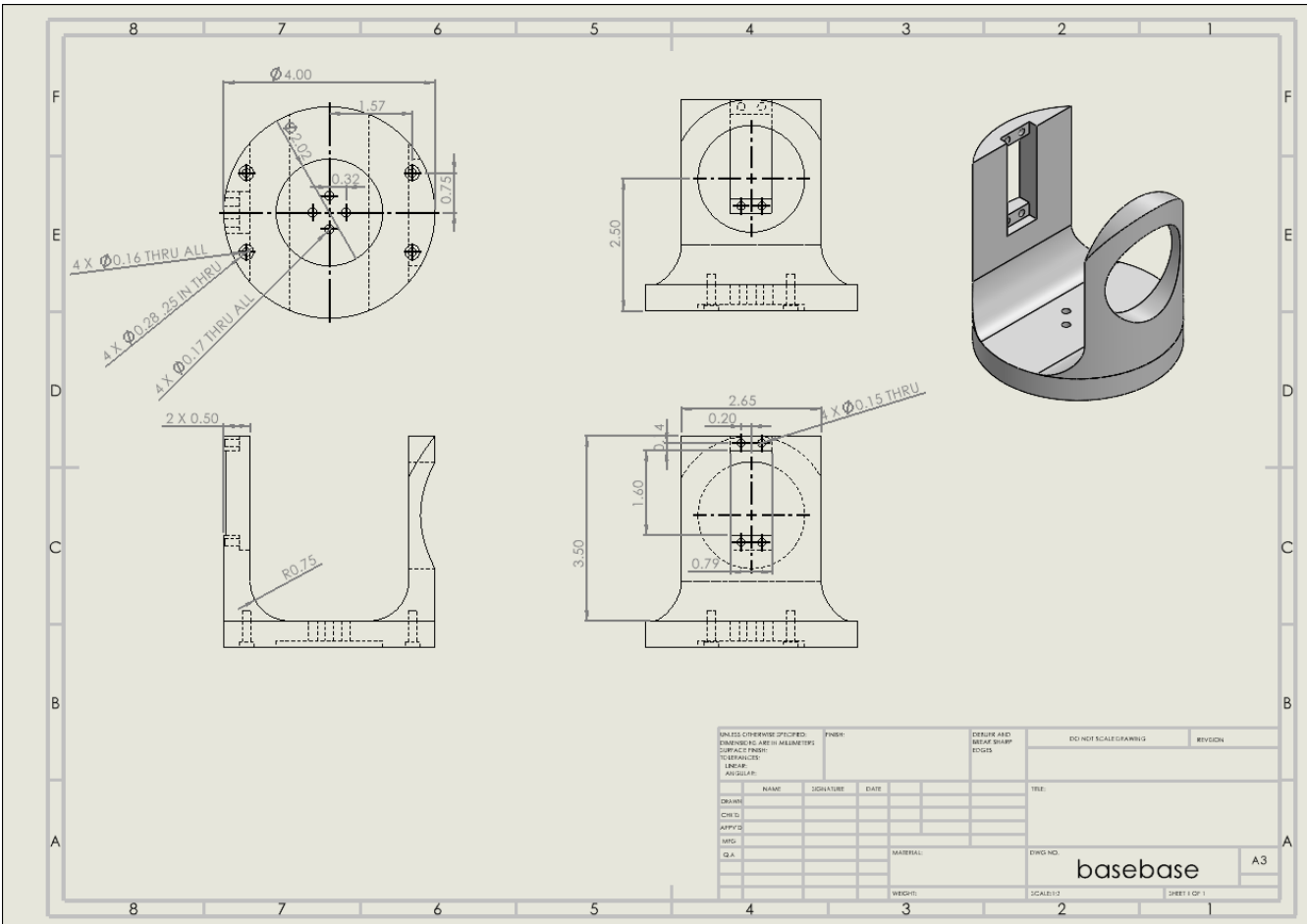


Figure 11: Shoulder Rotation Assembly Draft. Exploded view and detail drawing of the shoulder joint mechanism. This figure illustrates the interface between the Link 1 output horn and the Link 2 input mount, including the specific clearance gaps required to prevent binding during pitch rotation. Critical torque-transfer surfaces are marked with surface finish symbols

4 Electrical Architecture

4.1 Component Selection

The electrical system centers around the Raspberry Pi Pico, which manages PWM signals for the servos and handles serial data intake.

- **Microcontroller:** Raspberry Pi Pico connected via USB-UART bridge
 - **Joint 1 (Base) :** Nema 17 Stepper Motor
 - **Joint 2 (Shoulder):** DS3240MG 40kg * cm Servo
 - **Joint 3 (Elbow):** DS3218 20kg * cm Servo
 - **Gripper:** DS3218 20kg * cm Servo
- **Power Management:** A dedicated 6.14V Regulator along with one 100 μ F capacitor before each Servo motor ensures clean power for the Stepper Motor and all Servo Motors while the Pi Pico is supplied with 5V USB. The wire harness was assembled this way to avoid burn outs and to assure consistent voltage and current across all modules.

4.2 Schematic Design

Wiring was designed KiCad. Key considerations included separating high-current servo lines from sensitive logic lines and establishing a common ground. (View Next Page)

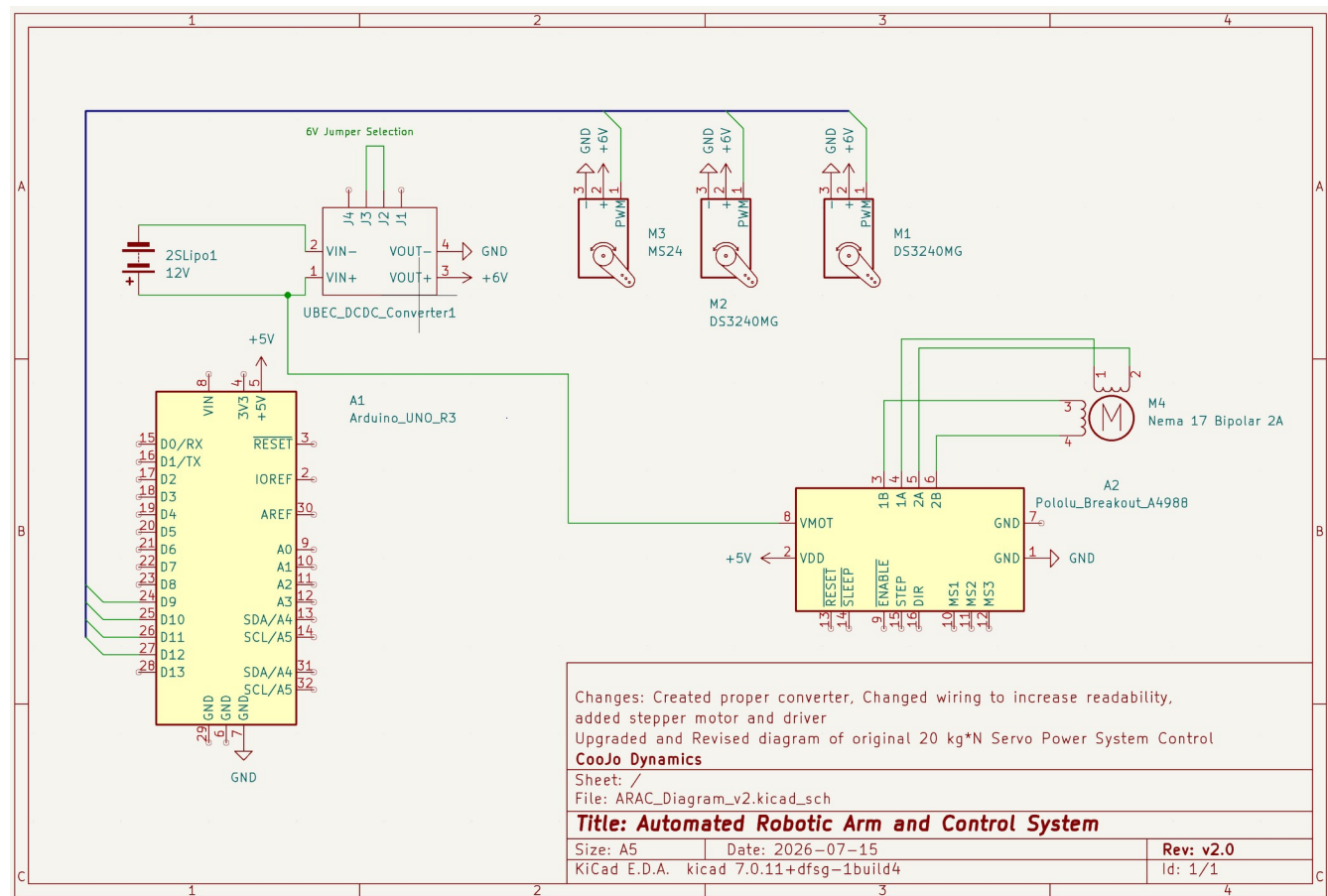


Figure 1: Electrical Schematic. Complete electrical schematic showing Pico pinout and servo connections.

5 Control System Implementations

5.1 Serial Communication Protocol

The arm accepts simple text commands via the Serial Monitor, allowing for easy testing and integration with higher-level software.

- **Command Format:** MOVE <base> <shoulder> <elbow>
- **Grip Command:** GRIP <state> (where state is 0 for open, 1 is closed)
- **Example:** MOVE 90 45 180 rotates the base 90°, shoulder to 45°, and elbow to 180°.

5.2 Firmware Logic

The firmware, written in C++, utilizes non-blocking loops to ensure smooth servo interpolation.

- **Initialization:** Servos are calibrated to neutral positions on boot.
- **Command Parsing:** Incoming serial strings are parsed to extract target angles.
- **Actuation:** Angles are converted to PWM duty cycles and sent to the GPIO pins

Below is a small code snippet of old code to explain how the movement works.

```
#include <Servo.h>

Servo myservo;
int angle = 0;

void setup() {
  myservo.attach(9); // Attach servo to Pin 9
  Serial.begin(9600); // Start serial communication
  Serial.println("Enter angle (0-95):");
}

void loop() {
  // Check if data is available from Serial Monitor
  if (Serial.available() > 0) {
    // Read the input angle
    angle = Serial.parseInt();

    // Constrain the angle between 0 and 95 degrees
    angle = constrain(angle, 0, 95);

    // Move the servo
    myservo.write(angle);

    // Confirm the action
    Serial.print("Moving to: ");
    Serial.print(angle);
    Serial.println(" degrees");

    // Prompt for next input
    Serial.println("Enter next angle (0-95):");
  }
}
```

6 Results and Performance

6.1 Testing Procedure

The physical prototype was subjected to a series of pick-and-place tests to validate Matlab simulations and mechanical design.

1. **Reach Test:** Verified the arm could reach the calculated workspace boundaries.
2. **Payload Test:** Confirmed the arm could lift up to approximately 830g at full extension without stalling.
3. **Repeatability Test:** Measured positional error over 50 consecutive cycles.

6.2 Performance Metrics

- **Positional Accuracy:** $\pm 8\text{mm}$ at maximum reach with a servo resolution of $\pm 0.5^\circ/\text{joint}$, below are the formulas used to find said accuracy

$$E_{max} \approx \sum_{i=1}^3 L_i \cdot \Delta \theta_i$$

$$A_p = \sqrt{(\bar{x} - x_c)^2 + (\bar{y} - y_c)^2 + (\bar{z} - z_c)^2}$$

- **Cycle Time:** 50ns
- **Max Grip Force:** The max grip force is equal to 20N (around 10N – 16N per gear) which was calculated using,

$$F = \tau / d$$

The physical testing confirmed the the simulation models accurately predicted the arm's torque requirements and workspace limitations

7 Conclusion

7.1 Verdict

This project successfully demonstrated that a low-cost, 3DOF robotic arm can be designed, simulated, and controlled with sufficient accuracy for light-duty pick-and-place operations. The integration of Matlab for kinematic validation prior to fabrication proved essential in identifying torque limitations early in the design process, preventing costly mechanical revisions. The system achieved a ± 8 mm positional accuracy at maximum reach and a 10–16 N grip force, confirming that the theoretical models accurately predict physical performance within the expected tolerance of hobby-grade servo actuators.

7.2 Objectives Check

All primary project objectives were achieved:

- **Mechanical Design:** The CAD model produced a functional 19-inch reach arm with an 830g payload capacity, meeting the target specifications.
- **Control Implementation:** The Raspberry Pi Pico successfully executed serial commands in real time, demonstrating reliable PWM generations for four servo actuators.
- **Simulation Validation:** Matlab simulations of forward and inverse kinematics matched physical test results within results within the calculated error bounds, validating the mathematical model.

7.3 Limitations

The conclusions of this project hold under the following boundary conditions:

- **Payload Constraint:** The 830g payload limit applies at the full extension (19 inches). Payload capacity increases to approximately 910g at mid-range (10 inches) due to reduced moment arm.
- **Static Loading:** Torque calculation assume static equilibrium. Dynamic loading (acceleration/deceleration) reduces effective payload by an estimated 20-30%.
- **Planar Workspace:** The 3DOF configuration provides a planar workspace. Full 3D spatial manipulation (excluding wrist orientation) is not possible without additional degrees of freedom.
- **Manual Commanding:** The current system requires manual serial input. Autonomous operations are not yet implemented.

7.4 Future Work

Based on the identified limitations, the following improvements are recommended for subsequent iterations:

1. **Computer Vision Integration:** Mounting a camera (e.g., Raspberry Pi Camera Module) to the end-effector or base would enable autonomous object detection using OpenCV, eliminating the need for manual coordinate input.
2. **Additional Degrees of Freedom:** Adding a wrist joint (4th DOF) would allow the arm to adjust the orientation of the gripper, significantly expanding the range of manipulable objects.
3. **Closed-Loop Control:** Implementing encoder feedback or force sensing on the gripper would allow the system to adjust grip force dynamically, preventing damage to delicate objects.
4. **Trajectory Planning:** Replacing discrete point-to-point movements with cubic spline interpolation in Matlab and the firmware would reduce mechanical stress and improve cycle time.

8 References

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